

Assignments → 35% → PSPx (Ass. section in End term)
End term → 40%
Quiz → 25%

Why Networks ??



What travels



Who are you talking to (IP Addresses)



How do we divide networks (Subnetting)



How do we find destination (Graph Algo)



How do routers send packets ??



How do humans use domains instead of
IPs



How do applications communicate (HTTP)



How to make comm. reliable



How do private networks work ??



Troubleshooting

Complete Journey.

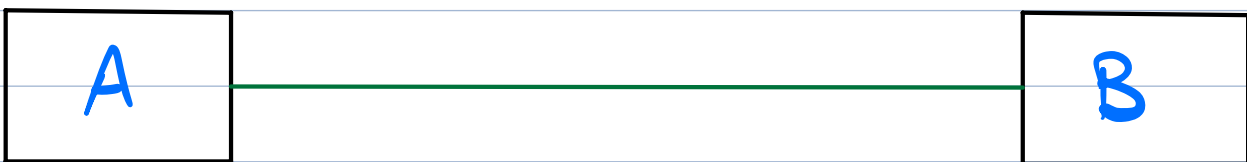
www.google.co.in.

- Sends a request to Google
- DNS
- Server responds with HTML.
- Routers.

Laptop → Wifi Router → ISP → Internet → Google Server

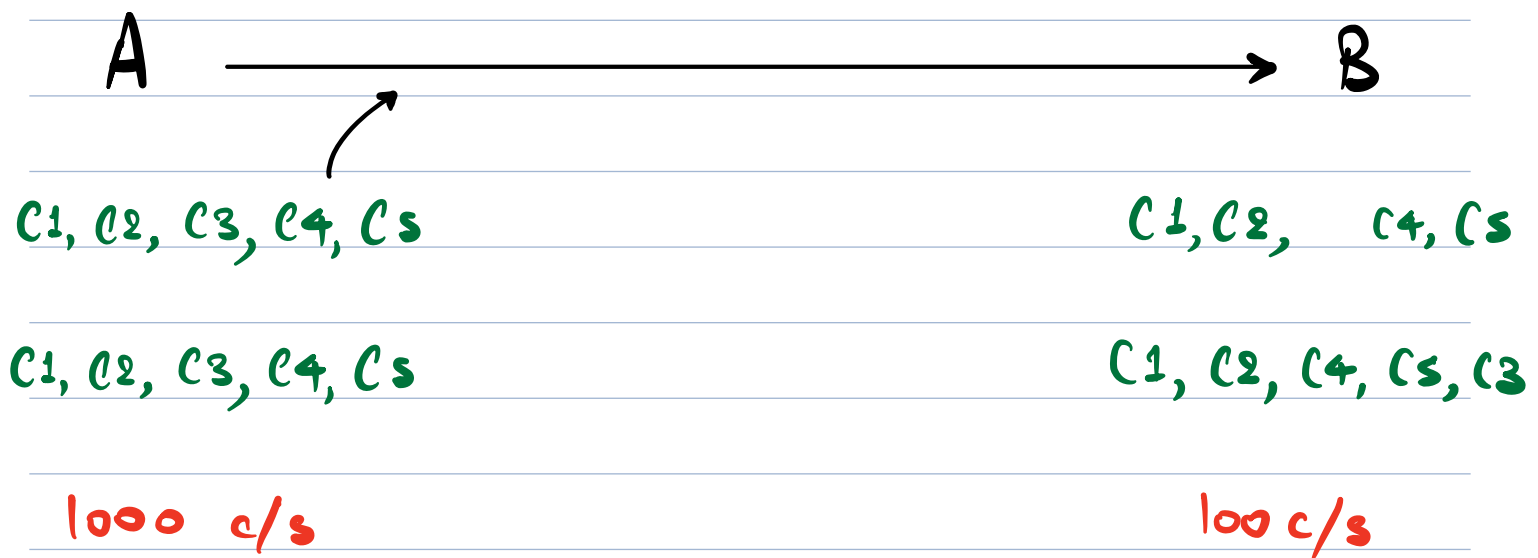
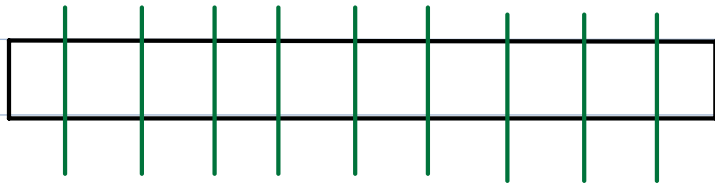
Laptop → Wifi Router → ISP → Router → Routers →
Routers → Google Network → Google Server

! Who owns the internet ??



→ Direct
→ Wifi

- 3) LAN
- 4) Multiple intermediate Networks.



Computer Network

⇒ Allows interconnected devices to communicate & exchange data / resources

Components of a Network

1) End Devices / Hosts

3) Links

3) Intermediate Devices

- Switch
- Router
- Access Points

Host → Inter. Devices → Host

4) Protocols.

HTTP

DNS

TCP

UDP

IP

3) Types of Networks

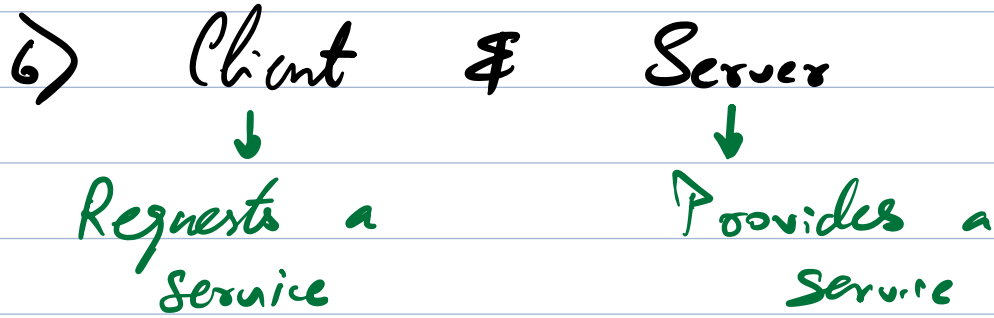
LAN - Local Area Network

WAN -

Internet

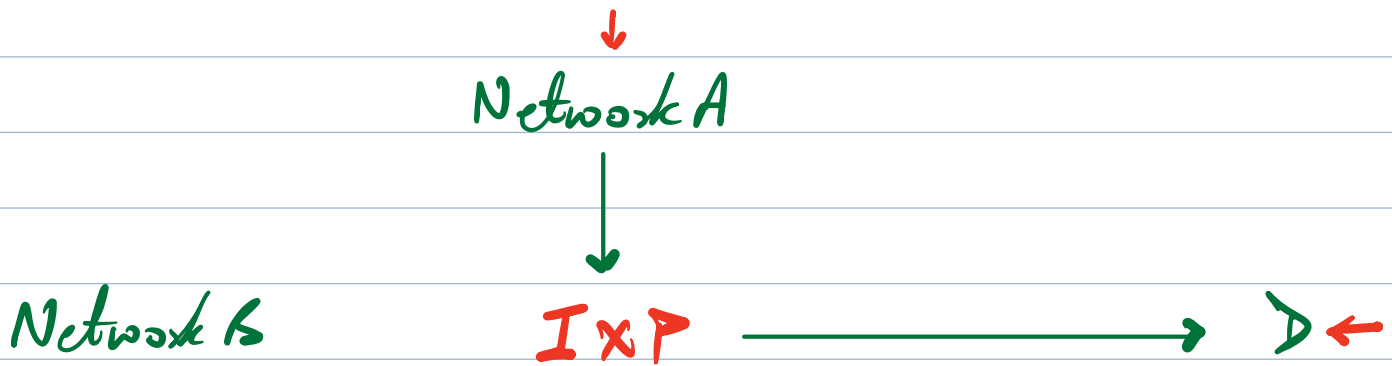
! Can internet be called as a very large LAN.

NO

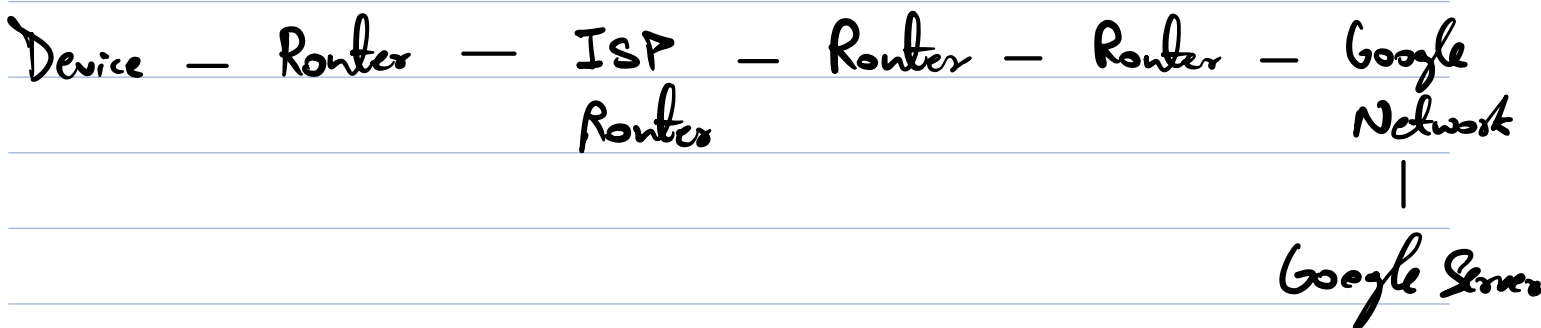


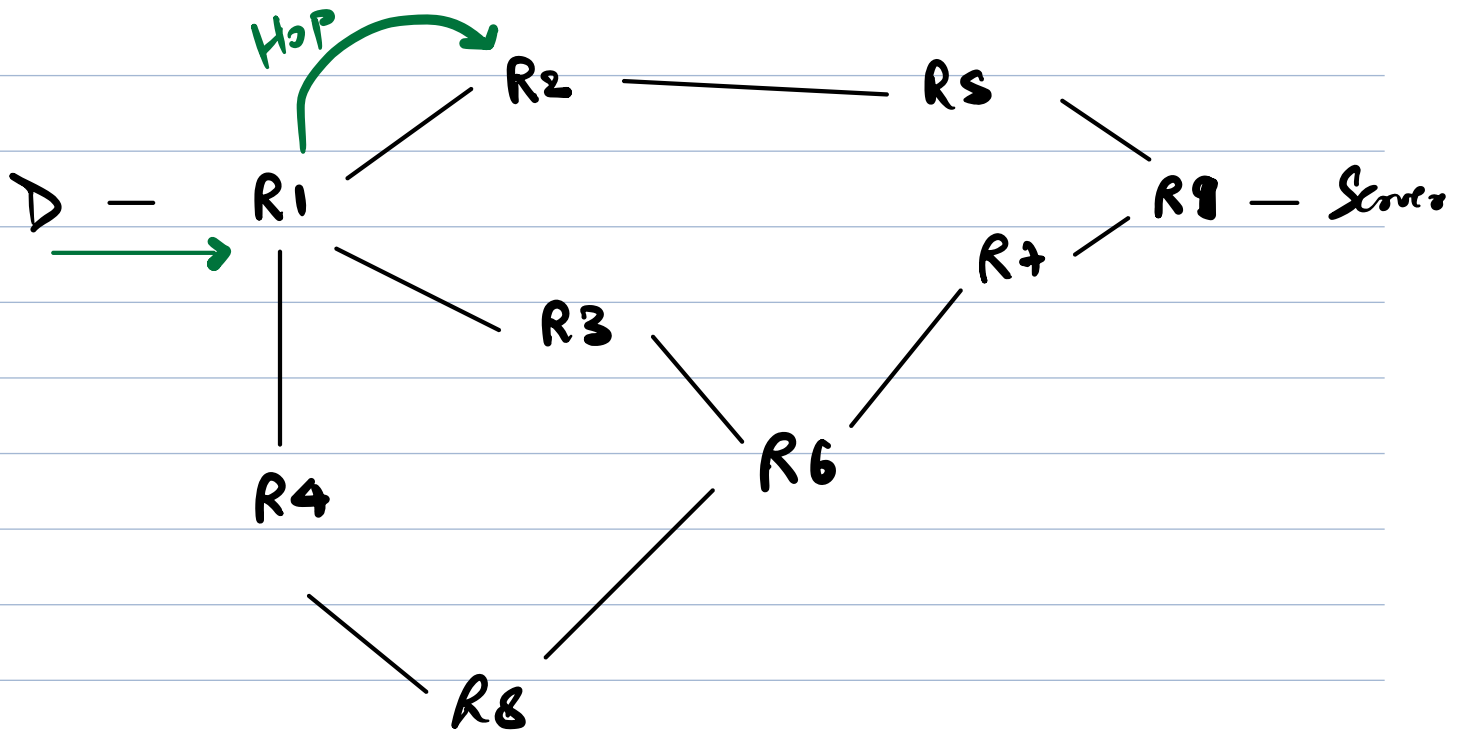
ISP

IXP (Internet Exchange Point)



C





* Physical Distance matters
 ↳ latency.

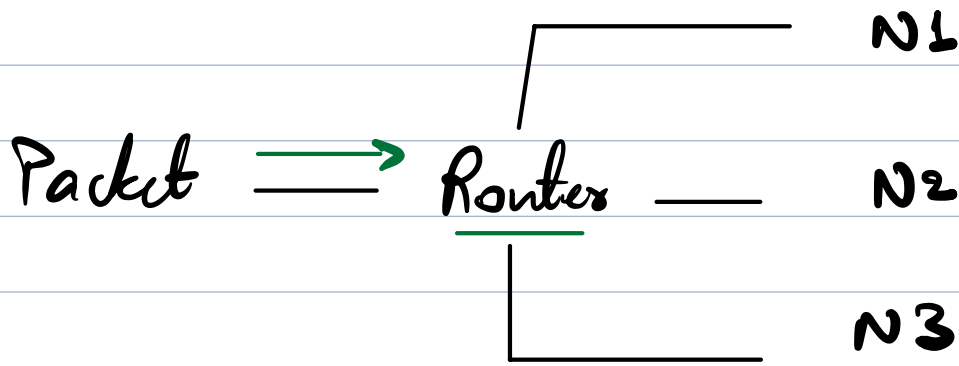
Network Edge v/s Network Core

↓
 End Points
 where communication
 starts / terminates

↓
 Intermediate
 devices

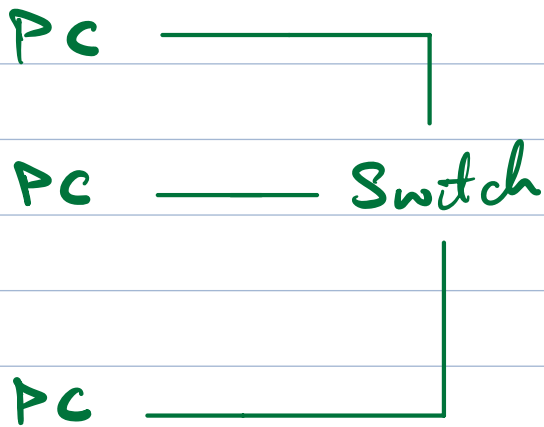
Routers

Devices



Switch

$\Rightarrow$ Connects devices within a LAN



Access Networks

Network Edge



Access Network



Network Core

R1 → R2 → R3 → R4



Access Network

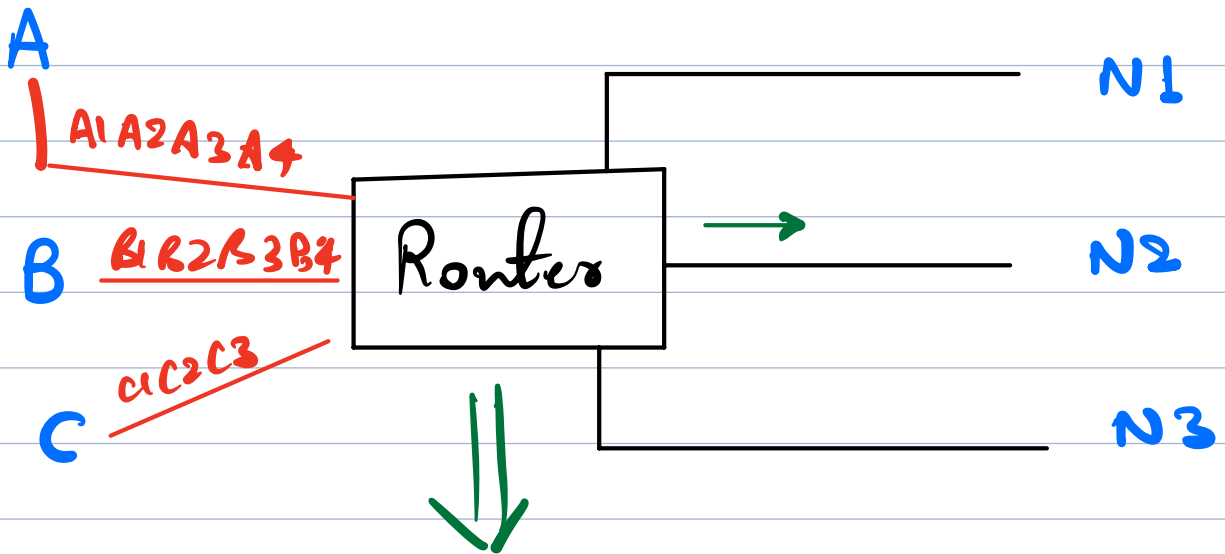


Network Edge

What is inside a Packet ??



Packet Switching vs Circuit Switching



A1, A2, B1, C1, B2, B3, C2, A3, C3, A4.